

Interim Project Report

Melbourne Airbnb Market Screening

Interim Project Report

CMCE30005 Business Analytics Challenge | Semester 2 2026 | TheNextChapter Group 2

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Repository: <https://github.com/Qihang-cpu/CMCE30005-TheNextChapter>

Introduction

Our client leases residential properties and runs them as Airbnb accommodation (“rental arbitrage”). With limited start-up funds, the client must choose which Melbourne areas and property types to investigate before signing leases and buying furniture; a poor choice ties up money and delays the business. Our project narrows this search.

We use the Inside Airbnb dataset supplied by the school, which records property details, locations and dated guest reviews. Because a review can only follow a completed stay, review counts are the only observable demand signal in these data. Locations are grouped by Local Government Area (LGA), a council area.

Our output is a shortlist of segments for further investigation, based on past review activity, plus a test of whether a model improves on segment averages. Financial assessment needs actual rent, furnishing and operating costs, which this dataset lacks.

Problem Definition and Objectives

Research question: Which types of Melbourne property — defined by council area (LGA), apartment or house, and number of bedrooms — most often reach a high level of guest-review activity, and does a statistical model add anything to simple segment averages when ranking them? “High” means at least 30 reviews in the year before the data were collected, the top quarter of a separate reference group of comparable listings.

Hypotheses: (H1) segments differ materially in the share of listings reaching that level; (H2) a model using listing attributes ranks listings better than the segment average alone.

Objectives: (1) a segment comparison table; (2) a model-versus-average test; (3) a shortlist with uncertainty ranges.

We study entire homes with one to three bedrooms, what a small operator can furnish and let whole; rental units and condos form the apartment group, homes and townhouses the house group. Each segment needs at least 50 listings, the smallest group whose share is stable within about ± 12 percentage points; prices of AUD 30–1,500 remove placeholder and luxury quotes; and a first review at least 365 days before collection gives a full year of exposure.

We call the share of listings reaching 30 reviews “attainment”; 30 reviews means at least 30 separate reviewed stays, roughly one every 12 days. Because stay length and review habits vary, attainment measures review activity, not bookings, occupancy or profit.

Data Description

The school dataset contains 25,728 listings with 90 variables, 9,390,720 calendar records and 1,026,690 reviews (Inside Airbnb, 2026), collected between 17 June and 1 July 2026. We use LGA, property type, bedrooms, capacity, bathrooms, amenities and review dates.

Cleaning converts price text to numbers, standardises missing values and extracts bathroom and amenity counts; price is missing for 6,553 listings and bedrooms for 4,679. The supplied last-12-months review field spans 366 dates, so we rebuilt counts over exactly 365 days from review dates (Table 1).

Table 1. Sample funnel

Stage	Listings
All listings in the dataset	25,728
Entire homes, 1–3 bedrooms	16,576
Standard dwelling types	14,895
Quoted price AUD 30–1,500	11,099
First review $\geq$ 365 days before collection	6,891
Analysis sample (50+ per segment, reference hosts excluded)	3,873
Reference group in the same segments	906

The reference group's 75th percentile is 29.75 reviews, giving a threshold of 30. The analysis sample (3,873 listings from 1,726 hosts) has 14 segments across six LGAs, about 64% in Melbourne LGA, and keeps 334 listings with zero annual reviews. Overall, 981 listings (25.33%) reach the threshold, and attainment ranges from 8.25% to 36.60% across segments (Figure 1).

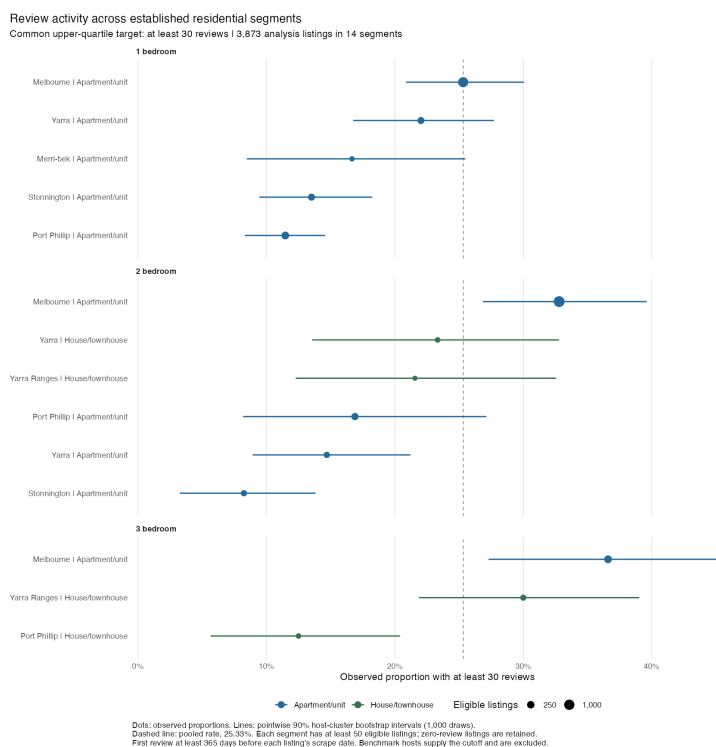


Figure 1. Share of listings reaching 30 reviews by segment; lines are 90% host-resampled ranges, dashed line the pooled 25.33% rate.

Widening the rules changes the population (17.51% attainment without the history rule, 18.29% without the price rule). Of listings without a usable price, 82% had no review in the past year, so

the priced sample is the more active part of the market. Only listings still active at collection are present, so observed attainment is likely higher than a new entrant would face.

Methodology and Analytical Approach

A fixed rule on host identifiers sets aside about 20% of hosts as the reference group that defines the 30-review target; they are excluded from all training and testing. We use five-fold validation: the data are split into five parts and each is tested in turn on a model trained on the other four, with each host's properties kept in one fold so a host with many near-identical apartments cannot appear in both training and test data (scikit-learn developers, n.d.). Missing values are imputed from training data only.

Logistic regression is the simple reference; random forest and gradient boosting capture nonlinear patterns. Basic features are location, bedroom-dwelling configuration, guest capacity, bathrooms and amenity count; one enhanced feature set adds coordinates, beds and nine specific amenities. On it we compare six variants: property-only logistic regression, random forest, gradient boosting and sigmoid-calibrated gradient boosting (which rescales predicted percentages), plus boosting with and without that step when current price and minimum stay are added. Review-based predictors, ratings and host badges are excluded.

Every model is compared with a segment-average baseline: each listing gets its segment's attainment among training hosts. AUC is how well the model orders active above inactive listings; 0.5 is a coin toss. Average precision is how many of its top picks really reach the target. Brier score is how far its percentages are from what happened; lower is better. R handles cleaning and tables; Python (scikit-learn 1.7.2) runs the models.

Current property details may have changed during the outcome year, so property-only models remain historical comparisons; model choice followed inspection of results and remains exploratory.

Analysis Plan and Progress to Date

Following Week 5 feedback, we dropped the earlier rent-versus-revenue (ROI) design, which needed rental data from outside the supplied dataset, and now compare only within the snapshot. Cleaning, review reconstruction, descriptive analysis and validation are complete.

Table 2. Predictive performance

Predictor	AUC	Average precision	Brier score
Segment average (training hosts)	0.581	0.285	0.1868
Basic property-only logistic	0.573	0.285	0.1886
Enhanced property-only random forest	0.640	0.350	0.1813

The enhanced random forest raises AUC by 0.059 and lowers Brier by 0.0055, beating the segment average in all five folds and all 20 further host splits (mean AUC 0.626, range 0.589–0.648); gradient boosting has a higher AUC (0.647) but a higher Brier. Adding current price and minimum stay lifts boosting to AUC 0.749 and Brier 0.162, but these are present-day settings, not attributes known before opening, so they are reported separately and excluded from the shortlist.

Table 3 ranks segments by forest predictions from models trained without each listing's host.

Table 3. Initial search candidates

Segment	Listings / hosts	Observed attainment	Mean predicted probability	95% range
Melbourne three-bedroom apartments	306 / 139	36.60%	34.45%	33.0–35.8%
Melbourne two-bedroom apartments	1,235 / 482	32.79%	31.41%	29.6–33.0%
Yarra Ranges three-bedroom houses/townhouses	90 / 77	30.00%	27.82%	26.1–29.3%

The ranges resample listings with predictions held fixed and exclude model-selection uncertainty. Melbourne three-bedroom apartments rank first in all 20 further splits and the top three recur in 19; Yarra Ranges rests on a smaller evidence base. Segment averages answer where to search; the property-level model answers whether a specific property found there is a good candidate — it changes no candidate here, so its value lies in scoring individual properties. The client can prioritise lease enquiries in these segments and verify permissions and costs before signing.

Remaining work:

1. Select the model within training data only and re-test it — Eric Huang (modelling lead), Week 9 (25 Sep).
2. Re-rank with uncertainty ranges under wider property-type and history rules — Maksym Xu (data lead), Week 10 (2 Oct).
3. One-page client brief with search priorities and a pre-lease checklist — Loc Le (report lead), Week 11 (by the 13 Oct presentation).
4. Final report and presentation — all, Weeks 11–12.

If the model does not beat segment averages after (1), the shortlist uses averages only; if rankings change under (2), we report a wider candidate set.

References

Inside Airbnb. (2026). Melbourne listings, calendar and reviews [June 2026 dataset supplied through CMCE30005 LMS].

Scikit-learn developers. (n.d.). Cross-validation and probability calibration. https://scikit-learn.org/stable/modules/cross_validation.html ; <https://scikit-learn.org/stable/modules/calibration.html>

OpenAI. (2026). ChatGPT/Codex assistance with project planning, report drafting and code review [Generative AI outputs, September 2026].

Anthropic. (2026). Claude Code assistance with analysis-code development [Generative AI outputs, September 2026].

AI Use Acknowledgement

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